2. TO FIND THE THICKNESS OF GIVEN IRON NAIL

Aim:

To find the thickness of the given iron nail.

Apparatus required:

Screw gauge and iron nail

Formula:

(i) Least Count (LC) = $\frac{\text{Pitch scale Reading}}{\text{No of divisions in the Head scale}}$

(ii) Thickness (t) = Pitch scale Reading (PSR) + Head scale coincidence (HSC) × Least Count (LC)] ± zero correction

$t = PSR + (HSC \times LC) \pm ZC$

Error:

(i) If positive error is 5 points, for zero correction, subtract 5 points.

$t = PSR + (HSC \times LC) - ZC$

$t = PSR + (HSC \times LC) - 5$

(ii) If negative error is 95 points, for zero correction add 5 points (100 -95 =5).

$t = PSR + (HSC \times LC) + ZC$

$t = PSR + (HSC \times LC) + 5$

(iii) If no correction is needed, $t = PSR + (HSC \times 0.01) \pm 0$

Procedure:

- The Least count of screw gauge is 0.01 mm.
- The zero error is to be found when the two faces of the screw gauge touch each other.
- Then place the iron nail between the two faces of the screw gauge. The pitch scale reading (PSR) and head scale coincidence (HSC) are to be noted.
- Repeat the process by placing other parts of the iron nail in the screw gauge.
- Tabulate the readings.

Zero corection: Least count: 0.01 mm

Sl. No.	Pitch Scale Reading PSR (mm)	Head Scale Coincidence (HSC)	Thickness of the iron nail t = PSR + (H S C × LC) ± ZC (mm)
1			
2			
3			

Average: _____ cm

Result: The diameter (Thickness) of the iron nail is _____ mm.

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